

Asynchronous Activity 1

Student Early Warning Tool Using KNIME

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Course / Section	CS0065 Intelligent Systems / AN43
Repository	Fernando-Ken-Dylen-CS0065-AN43
Project folder	Technical-Assessment-01/DemoEarlyWarningTool

1. Completed KNIME workflow and model result

The screenshot displays the KNIME Analytics Platform interface. The main workspace shows a workflow starting with a CSV Reader, followed by Column Filter, Normalizer, and Table Partitioner. The data is then split into three parallel paths for Decision Tree, Logistic Regression, and Random Forest models, each with a corresponding learner and predictor node. The outputs are compared using Scorer nodes. A zoomed-in view of a Scorer node is shown on the right, with the first column set to 'risk_status' and the second column to 'Prediction (risk_status)'. The sorting strategy is 'Insertion order'. The missing values are set to 'Ignore'.

Scorer

First column: risk_status

Second column: Prediction (risk_status)

Sorting strategy: Insertion order

Reverse order: ☐

Use name prefix: ☐

Missing values: Ignore Fail

Table

Rows: 2 | Columns: 2 | Sorting | Filter

#	RowID	Not At Risk	At Risk
1	Not At Risk	5	0
2	At Risk	0	4

The workflow reads and prepares the student dataset, partitions it, and compares Decision Tree, Logistic Regression, and Random Forest models. The visible Scorer result shows 5 Not At Risk and 4 At Risk records classified correctly.

2. Required Demo project files

DemoEarlyWarningTool - required deliverables

D:\github\Fernando-Ken-Dylen-CS0065-AN43\Technical-Assessment-01\DemoEarlyWarningTool

Name	Type	Size
DemoEarlyWarningTool.knwf	KNIME workflow	126.0 KB
README.md	Markdown document	2.2 KB
student_performance_knime.csv	CSV dataset	0.9 KB

Prepared locally; Git commit and public upload are pending user confirmation.

The DemoEarlyWarningTool folder contains the exported .knwf workflow, the required README.md file, and the dataset used by the CSV Reader. The final evidence PDF is added after public-repository verification.

3. Repository organization

Repository layout prepared for Asynchronous Activity 1

D:\github\Fernando-Ken-Dylen-CS0065-AN43

```
Fernando-Ken-Dylen-CS0065-AN43/
|-- .gitignore
|-- README.md
`-- Technical-Assessment-01/
    |-- README.md
    |-- DemoEarlyWarningTool/
        |-- DemoEarlyWarningTool.knwf
        |-- README.md
        |-- student_performance_knime.csv
        |-- Fernando_Ken_Dylen_KNIME_GitHub_Evidence.pdf (to be finalized)
    |-- Python and Streamlit implementation files

Git status: local changes prepared; no commit or push performed yet.
```

The activity is organized under Technical-Assessment-01 while preserving the required DemoEarlyWarningTool project folder and its deliverables.

4. Public GitHub repository

This page will be replaced with the final GitHub repository screenshot after the user confirms publishing the prepared files.

5. Public-access verification

This page will be replaced with an anonymous/private-session verification screenshot after the repository is published and set to Public.

Verification checklist

- Repository link opens and visibility is Public.
- DemoEarlyWarningTool folder is visible.
- DemoEarlyWarningTool.knwf uses the .knwf extension.
- README.md is present and documents the project, models, and run instructions.
- student_performance_knime.csv is included and can be selected in CSV Reader.
- Evidence PDF opens from the GitHub repository.
- Workflow archive contains KNIME workflow metadata and node settings.